

WEEK 1 · LAB

Lab 1: Your First Instance, and Getting Rid of It

Create an account, set a budget alert before anything else, run one machine, and prove you removed it completely.

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DURATION

One 2-hour block

MARKS

10

SUBMIT

icclab1-<ROLLNO>.pdf

OFFERING

Odd Semester 2026-27

Before you create anything

PITFALL · SET A BUDGET ALERT FIRST. NOT LATER.

The most common way a student on this course incurs a real bill is leaving a load balancer, a managed database or a GPU instance running over a holiday. Cloud providers do not stop you. A budget alert does.

Do this before you create a single resource, and screenshot the confirmation.

0.1 Create a free-tier account with one provider. Set a spending alert at a small amount. Enable multi-factor authentication on the root or owner account.

0.2 Create a **non-root** user or service principal with only the permissions you need, and use it for everything afterwards. Explain in two sentences why working as root is a bad habit even when it is convenient.

Part 1: One machine (3 marks)

1.1 Launch the smallest available Linux instance. Record: region, instance type, vCPUs, memory, hourly price.

1.2 Connect to it over SSH using a key pair you generated. Do not use a password.

1.3 Run `lscpu`, `free -h`, `df -h`, and a quick `sysbench` or a loop that measures rough CPU throughput. Record the results.

1.4 From your own machine, measure the round-trip latency to the instance (`ping` or `mtr`). Then look up the great-circle distance to that region and compute the theoretical minimum round trip at the speed of light in fibre (about 200 000 km/s). Compare. How much of the latency is physics and how much is everything else?

Part 2: Regions are not interchangeable (2 marks)

2.1 Launch, or simply measure latency to, an endpoint in three regions: one near you, one on another continent, one in between.

2.2 Tabulate latency against distance. Comment on what this means for an application with a strict interactive latency budget, and connect it to why edge computing exists.

Part 3: Storage (2 marks)

3.1 Create an object storage bucket. Upload a file with the CLI. Download it again. Record the commands.

3.2 Make it public, confirm you can fetch it without credentials, then **make it private again**. Explain in three sentences why this two-minute exercise is the single most common cause of real-world cloud data breaches.

3.3 Look up the price per GB-month for the standard, infrequent-access and archive tiers of your provider, and the retrieval cost of each. Say which you would use for: a dataset you read daily, last year's experiment logs, and a legal retention copy.

Part 4: Teardown, and proof (3 marks)

4.1 Delete everything: instance, disks, bucket, IP addresses, key pairs, security groups. Disks and static IP addresses commonly survive instance deletion and keep billing.

4.2 Prove it. Run the CLI list command for each resource type and paste the empty output. A screenshot of a console page is not proof.

4.3 Check the billing or cost-explorer page and report the total incurred.

NOTE · TEARDOWN IS MARKED IN EVERY LAB ON THIS COURSE

Provisioning something is half the skill. Being able to remove it completely, and to demonstrate that nothing was left behind, is the half that keeps you employable.

What to submit

`icclab1-<ROLLN0>.pdf` : the budget alert confirmation, your instance details, the latency-against-distance table with the speed-of-light comparison, the storage tier recommendations, and the teardown evidence as CLI output.

MARKS	FOR
2	Part 0: budget alert, MFA, non-root working identity
3	Part 1, especially the latency physics calculation
2	Part 2: the regional comparison and its interpretation
2	Part 3: storage tiers chosen with reasons
1	Part 4: complete teardown proved with CLI output